

DAVIDSON COUNTY AP, NC, USA

WMO: 722079

Lat: 35.781N

Lon: 80.304W

Elev: 733

StdP: 14.31

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 53888

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.4	21.8	0.2	5.7	25.1	5.0	7.3	27.9	16.4	54.2	14.2	50.4	1.3	20	0.350

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	91.5	73.9	90.3	73.9	88.0	73.4	77.8	85.6	76.9	84.7	75.9	83.5	3.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.7	138.3	80.3	74.7	133.7	79.6	73.5	128.0	78.7	41.7	85.8	40.7	84.9	39.8	83.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.9	11.0	9.6	10.4	95.7	6.1	2.7	6.0	97.6	2.5	99.2	-0.9	100.8	-5.4	102.8		
(5)			9.0	80.2	5.5	3.8	5.1	82.9	1.9	85.1	-1.2	87.2	-5.2	89.9	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	59.8	40.3	43.4	51.2	60.0	67.6	75.0	77.6	76.5	71.3	60.4	49.1	43.8
	DBStd	15.24	10.21	9.55	9.47	7.82	6.51	4.47	3.77	4.25	5.68	7.98	8.19	8.86
	HDD50	1034	328	229	106	12	0	0	0	0	0	11	113	235
	HDD65	3385	766	606	436	187	53	2	0	0	15	185	478	657
	CDD50	4600	28	44	142	312	547	749	854	821	638	334	87	44
	CDD65	1474	0	1	7	37	134	301	389	356	203	43	2	1
	CDH74	12130	0	7	78	397	1063	2524	3341	2858	1488	346	23	5
	CDH80	4181	0	0	7	68	267	899	1330	1054	489	66	1	0
Wind	WSAvg	2.7	2.9	3.5	3.8	3.9	3.0	2.3	2.0	1.9	2.0	2.1	2.3	2.3
Precipitation	PrecAvg	43.6	3.7	3.1	4.3	3.7	3.3	4.1	4.3	3.8	3.9	3.3	3.0	3.3
	PrecMax	62.4	7.0	5.4	8.3	8.3	8.9	8.1	7.5	7.4	9.3	13.3	6.8	6.6
	PrecMin	29.8	1.1	1.0	1.3	0.5	0.5	1.7	1.9	1.3	0.4	0.0	0.4	1.2
	PrecStd	7.0	1.5	1.1	1.8	1.9	1.9	1.9	1.7	1.5	2.0	2.9	1.6	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.3	74.3	80.9	84.5	89.8	93.2	94.9	96.8	91.6	86.1	78.6	72.8
		MCWB	60.4	62.8	60.4	63.7	71.4	72.8	75.1	73.2	70.8	72.0	63.6	61.9
	2%	DB	64.5	68.8	75.4	81.8	85.9	90.6	91.4	91.0	89.3	81.3	72.6	67.2
		MCWB	57.4	58.5	59.2	62.7	69.6	73.2	75.3	74.5	71.9	67.2	61.3	60.7
	5%	DB	61.3	64.1	72.2	79.1	82.6	88.2	90.0	88.5	85.9	78.5	68.0	63.2
		MCWB	54.7	55.7	58.4	62.8	68.4	72.7	75.0	74.7	72.2	66.3	59.5	57.5
	10%	DB	57.0	60.6	67.8	75.2	80.8	85.4	87.5	85.8	82.2	73.4	64.0	59.1
		MCWB	50.8	51.8	56.9	60.9	67.4	72.4	74.3	74.0	69.7	65.5	56.6	53.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.1	65.3	66.0	70.7	74.8	77.5	79.2	79.1	77.2	75.2	67.9	66.6
		MCDB	66.6	70.9	72.1	76.2	80.9	86.3	86.7	85.7	83.7	79.8	71.7	69.3
	2%	WB	59.8	62.1	63.4	68.0	72.5	76.2	77.9	77.7	75.4	72.8	64.6	63.0
		MCDB	63.3	65.9	71.2	74.4	80.5	84.9	86.3	85.0	81.8	76.7	69.0	65.5
	5%	WB	56.3	58.2	61.0	65.8	71.0	75.2	77.0	76.8	74.0	70.0	61.6	59.3
		MCDB	60.3	63.0	68.2	73.9	79.2	83.6	85.5	83.9	80.1	74.8	65.9	61.9
	10%	WB	52.2	52.9	58.8	63.8	69.5	74.1	76.0	75.6	72.6	67.3	58.3	54.6
		MCDB	56.4	57.9	66.2	71.9	76.8	81.9	84.0	82.6	78.4	72.8	63.4	58.2
Mean Daily Temperature Range	5% DB	MDBR	19.8	20.3	22.1	22.8	20.7	19.1	18.6	18.1	19.4	21.8	22.8	19.8
		MCDBR	24.4	25.1	26.3	26.9	23.8	22.5	21.5	21.3	24.2	25.2	27.5	23.7
		MCWBR	17.4	16.2	13.5	11.8	9.8	8.2	7.3	7.6	8.9	12.8	16.7	16.8
	5% WB	MCDBR	20.2	21.6	21.9	20.5	19.3	18.6	18.3	17.5	18.4	18.4	20.5	18.6
		MCWBR	17.5	16.6	13.2	10.7	8.8	8.0	7.4	7.6	8.4	10.9	16.2	15.9
Clear-Sky Solar Irradiance	taub	0.311	0.324	0.358	0.398	0.447	0.480	0.529	0.512	0.421	0.379	0.327	0.310	
	taud	2.496	2.455	2.395	2.287	2.183	2.126	2.014	2.044	2.295	2.386	2.491	2.532	
	Ebn at Noon	283	291	289	281	267	258	244	245	265	267	274	276	
	Edh at Noon	26	30	36	42	47	49	55	52	39	32	26	24	
All-Sky Solar Radiation	RadAvg	815	1022	1360	1702	1888	2025	1889	1724	1449	1185	900	698	
	RadStd	53	125	95	133	154	147	124	83	130	151	73	62	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (63 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+252	+176

Nomenclature: See separate page